



UITS

**UNIVERSITY OF INFORMATION
TECHNOLOGY AND SCIENCES**

PROJECT PROPOSAL

For

UniQuery — *“Ask. Explore. Choose.”*

A Smart University Information and Admission Assistant

Prepared By:

Israt Jahan — 0432320005101130

Md. Mehedi Hasan — 0432320005101082

Course: Software Project Design & Development (CSE 416)

Academic Session: Autumn 2026

Prepared For:

Dr. Mahfida Amjad Dipa

Senior Teacher

Department of CSE

❑ INTRODUCTION

Selecting an appropriate university and academic program is one of the most critical career decisions for high school and secondary education graduates. In Bangladesh, high school graduates face significant complexity during admission cycles across various public and private institutions.

Information dissemination in the higher education ecosystem is heavily fragmented across static university websites, unstructured PDF circulars, newspaper notices, and social media channels. Students must spend considerable time interpreting varied eligibility rules, exam board conversion formulas, tuition fee structures, and strict application schedules. Consequently, applicants frequently rely on informal advice or costly admission coaching centers, leading to financial stress, reliance on misinformation, and missed deadlines.

UniQuery bridges this gap by acting as a centralized, interactive, AI-driven academic counselor. Built on an intuitive **"Ask, Explore, Choose"** framework, UniQuery leverages modern conversational AI, real-time database lookups, and visual comparison tools to make university discovery seamless, accessible, and personalized for every student.

❑ PROBLEM STATEMENT

Admission-seeking students lack a unified, intelligent, and real-time guidance platform to search, evaluate, and compare university options based on their academic qualifications, budget constraints, and career goals.

The primary challenges faced by applicants include:

- **Information Fragmentation:** Admission circulars, subject requirements, seat counts, and tuition fees are scattered across disparate websites and social media pages.
- **Complexity in Eligibility Assessment:** Manual calculation of eligibility across diverse HSC/SSC GPA criteria, unit requirements, and background subject rules is confusing and prone to errors.
- **Time-Consuming University Comparison:** Comparing multiple universities regarding tuition fees, available waiver schemes, lab infrastructure, and faculty credentials requires dozens of hours of repetitive manual research.
- **Overwhelmed Support Systems & Missed Deadlines:** Official admission helpdesks are often unreachable during peak cycles, leaving critical inquiries unaddressed and causing students to miss key application deadlines.

- **Dependence on Unofficial Advice:** Unverified posts in informal social media groups often spread outdated or inaccurate admission rules.

❏ PROPOSED SOLUTION

UniQuery addresses these issues by delivering a web-based platform with an integrated conversational AI chatbot. The system operates through a structured three-stage concept:

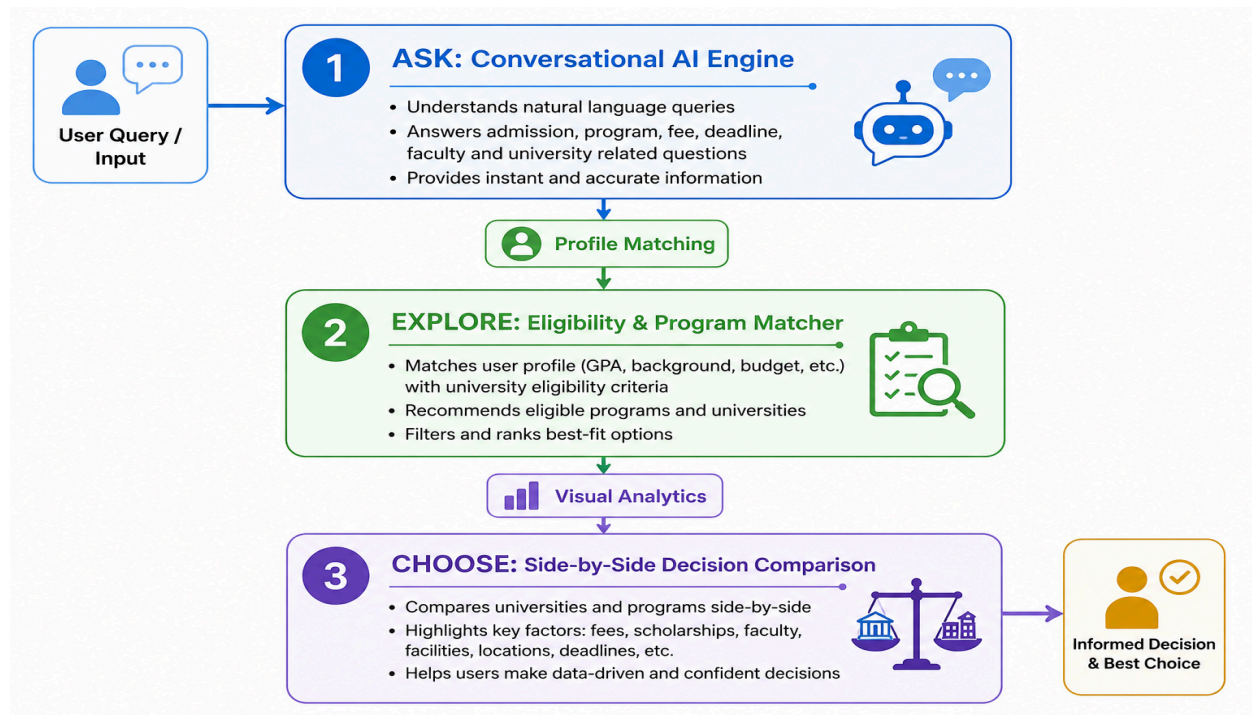


Figure: UniQuery Three-Stage Core Concept Workflow (Ask, Explore, Choose)

- **ASK (Natural Language Interaction):** Students interact with an AI chatbot to ask free-form questions regarding admission criteria, department details, seat counts, and application deadlines.
- **EXPLORE (Eligibility & Discovery):** Students input their SSC/HSC GPAs, academic group, and budget preferences to dynamically filter and view suitable academic programs.
- **CHOOSE (Data-Driven Decision Making):** An interactive comparison dashboard allows users to compare shortlisted universities side-by-side across tuition fees, waiver requirements, campus amenities, and deadlines.

❑ OBJECTIVES

The primary academic and technical objectives of the UniQuery project are:

1. **Centralize Admission Information:** Aggregate public and private university admission data into a structured single-source platform.
2. **Provide Conversational Guidance:** Implement a Natural Language Processing (NLP) conversational AI chatbot capable of instantly answering natural-language queries regarding admission rules, tuition fees, and deadlines.
3. **Automate Eligibility Matching:** Develop an automated eligibility rule engine that evaluates student GPAs and background groups to instantly display qualified programs.
4. **Enable Side-by-Side Comparison:** Build an interactive comparison dashboard to compare multiple universities based on tuition costs, financial aid, faculty ratings, and lab facilities.
5. **Track Critical Deadlines:** Incorporate a centralized notice and deadline tracking system to display real-time updates on circular releases, exam dates, and application links.
6. **Reduce Manual Research Overhead:** Streamline the university selection workflow to significantly decrease the time and effort required by applicants during admission seasons.

❑ SCOPE

In-Scope:

- **Targeted University Dataset:** Structured database covering prominent public and private universities in Bangladesh.
- **Conversational AI Chatbot:** AI-powered query handling for general admission inquiries, fee queries, and course descriptions.
- **Rule-Based Eligibility Matcher:** Automatic profile matching against GPA criteria, background science/arts/commerce groups, and unit constraints.
- **Comparison Engine:** Side-by-side visual and tabular metrics comparison (tuition fees, lab facilities, scholarships).
- **Notice Tracker:** Centralized display board with application links, active circular updates, and application deadlines.
- **Responsive Web Platform:** Accessible web interface tailored for both desktop and mobile web browsers.

Out-of-Scope:

- **Direct In-App Application Submission:** The platform provides official application portal links but does not process university application forms or admission fee payments directly.
- **Direct Integration with University Internal Databases:** Data will be managed via dedicated platform administrative modules rather than real-time integration into external legacy systems.
- **Seat Plan & Roll Number Tracking:** Individual exam hall seat allocation and roll numbers will remain managed by official university portals.
- **Native Mobile Applications:** The prototype will be developed as a responsive web application; native iOS/Android builds are reserved for future phases.

❑ TARGET USERS

- **Admission-Seeking Students:** HSC graduates, A-Level students, and diploma holders searching for suitable undergraduate programs.
- **Parents and Guardians:** Individuals evaluating tuition costs, scholarship eligibility, campus safety, and infrastructure to assist in financial planning.
- **Academic Counselors:** High school guidance advisors using structured comparison tools to assist students.
- **System Administrator:** Internal role responsible for maintaining, verifying, and updating university circular data, deadlines, and system parameters.

❑ FUNCTIONAL REQUIREMENTS

Conversational AI Chatbot (Ask Module)

- **Natural Language Querying:** Answers free-form queries regarding admission rules, tuition fees, faculty lists, and campus facilities in real time.
- **Context Awareness:** Maintains conversational state to handle follow-up questions effectively.

Smart Eligibility Matcher (Explore Module)

- **GPA & Group Input:** Accepts SSC/HSC GPAs, subject-wise grades, and academic backgrounds (Science, Business Studies, Humanities).
- **Rule Engine:** Evaluates student profiles against university-specific eligibility rules and displays matching eligible programs.

Side-by-Side University Comparison (Choose Module)

- **Multi-Parametric Comparison:** Displays comparative visual tables across tuition fees, waiver options, lab facilities, rankings, and locations.
- **Dynamic Filtering:** Filters institutions based on maximum budget limits, preferred geographic region, or university type (Public/Private).

Centralized Notice & Deadline Tracker

- **Schedule Dashboard:** Aggregates active application deadlines, exam schedules, and circular publication dates.
- **Application Links:** Provides direct links to official university admission application portals.

University & Program Profile Information

- **Detailed Institutional Profiles:** Displays accredited faculty details, laboratory capabilities, student-to-faculty ratios, and campus amenities.

Tuition Fee and Scholarship Matrix

- **Fee Breakdown:** Itemized total cost breakdowns across semesters/years.
- **Scholarship Rule Calculator:** Displays available merit-based waivers based on HSC/SSC result combinations.

❑ NON-FUNCTIONAL REQUIREMENTS

- **Usability:** Minimalist UI requiring minimal clicks to access critical eligibility and comparison features.
- **Performance:** AI chatbot response generation latency targeted under 2.5 seconds; page load time under 2.0 seconds.
- **Availability:** Operational uptime target of 99.5% on cloud-hosted prototype server infrastructure.
- **Security:** Standard SSL/TLS encryption for transit data security; sanitized inputs to protect against web vulnerabilities (XSS, SQLi).
- **Scalability:** Modern modular architecture capable of accommodating continuous expansion of university records.
- **Responsiveness:** Fluid grid rendering across mobile, tablet, and desktop display viewports.

PROPOSED METHODOLOGY

UniQuery uses a structured execution flow from user entry to final university selection:

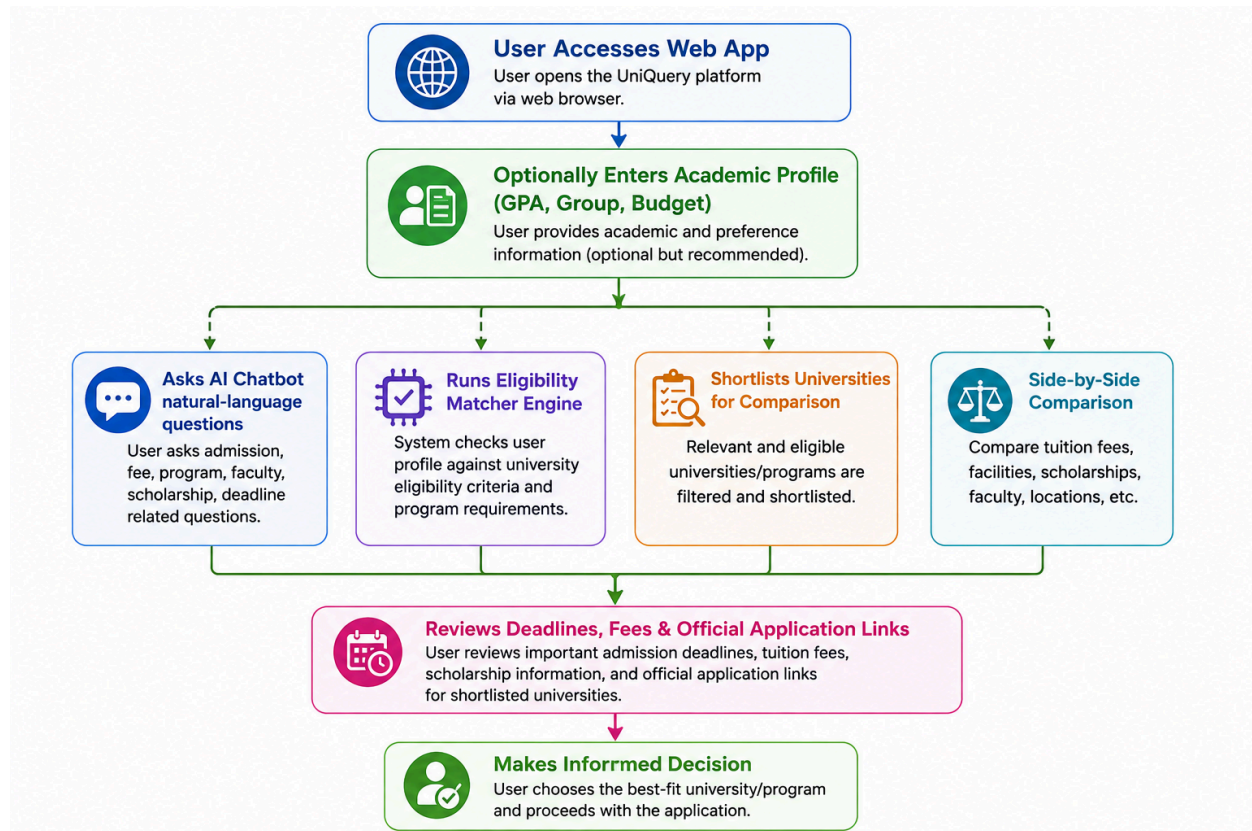


Figure: UniQuery End-to-End System Methodology and User Process Flow

1. **User Access & Profile Input:** The applicant accesses the UniQuery platform and inputs relevant academic profile information (SSC/HSC GPA, group, and budget preferences).
2. **Conversational AI Chatbot & Eligibility Processing:** The user asks queries via the AI chatbot or initiates the eligibility engine.
3. **Data Retrieval & Program Matching:** The backend processes queries, checks stored eligibility rules, and generates eligible program lists.
4. **Side-by-Side Comparative Analysis:** The student compares shortlisted programs across tuition fees, scholarships, and campus facilities.
5. **Notice Tracking & Portal Redirection:** The user reviews active admission schedules and accesses direct application links to complete their admission application on official portals.

❑ FEASIBILITY ANALYSIS

Technical Feasibility

The project is being developed by a 2-member student engineering team with distinct domain responsibilities:

- **Israt Jahan:** Handles system architecture, AI chatbot integration, NLP logic, frontend UI/UX design, and backend API integration.
- **Md. Mehedi Hasan:** Manages relational database design, data gathering and normalization, database maintenance, and comprehensive system documentation.

The required technical tooling (web frameworks, relational database, LLM API integration) is standard, well-documented, and achievable within the technical capacity of a 2-student academic team.

Economic Feasibility

The platform heavily utilizes open-source software, free cloud tiers, and pay-as-you-go AI API models. As demonstrated in the detailed Budget section, the financial expenditure required to build and test this prototype is minimal, highly realistic, and fully economically feasible.

Operational Feasibility

UniQuery directly solves established pain points for high school graduates and parents. The natural language conversational AI chatbot eliminates complex navigation, ensuring high adoption rates and ease of use for non-technical users.

Schedule Feasibility

The proposed 14-week timeline provides adequate buffer periods across project initiation, architecture design, feature implementation, testing, and final deployment.

❑ WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) for **UniQuery** follows a hierarchical decomposition model structured up to **3 Levels** to ensure systemic planning, execution, and tracking. The top-level goal (Level 1) is divided into five core operational phases (Level 2), which are further decomposed into actionable technical sub-tasks (Level 3).

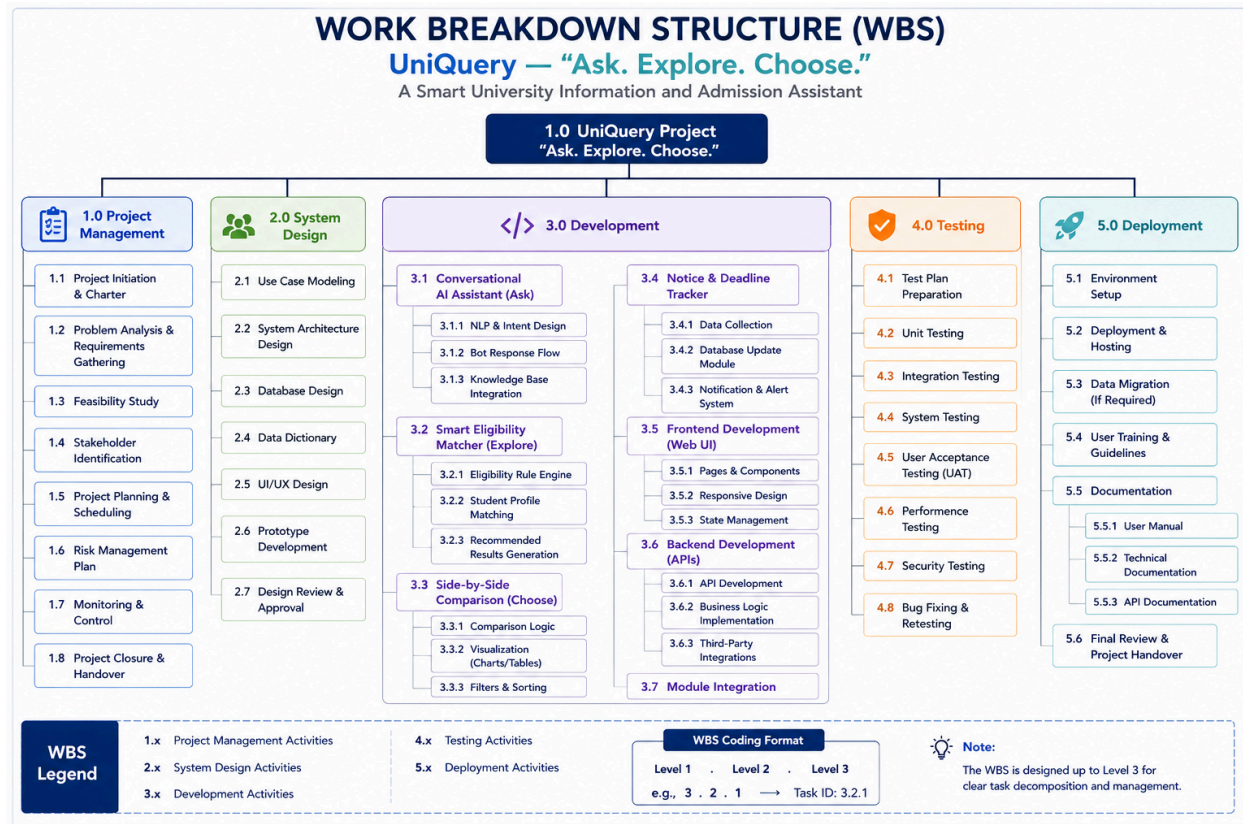


Figure: UniQuery Hierarchical Work Breakdown Structure (WBS)

Level 1: Overall System Goal

- **1.0 UniQuery Project (“Ask. Explore. Choose.”):** Represents the primary deliverable—an AI-powered, centralized university information and admission platform.

Level 2 & Level 3: Phase-Wise Task Decomposition

1.0 Project Management

Establishes the administrative foundation, project boundary, and execution plan.

- **Initialization & Scope:** Involves drafting the *Project Initiation & Charter* (1.1), conducting *Problem Analysis & Requirements Gathering* (1.2), and performing a comprehensive *Feasibility Study* (1.3).
- **Planning & Control:** Encompasses *Stakeholder Identification* (1.4), *Project Planning & Scheduling* (1.5), and drafting a proactive *Risk Management Plan* (1.6).

- **Execution Monitoring:** Continuous oversight via *Monitoring & Control (1.7)* leading to *Project Closure & Handover (1.8)* upon completion.

2.0 System Design

Focuses on converting requirements into technical blueprints, database schemas, and interface mockups before coding.

- **Architectural Modeling:** Directs *Use Case Modeling (2.1)* to define system boundaries and *System Architecture Design (2.2)* for frontend-backend interaction.
- **Data & Interface Engineering:** Covers relational schema construction via *Database Design (2.3)*, establishing a *Data Dictionary (2.4)*, creating interactive visual layouts through *UI/UX Design (2.5)*, developing an initial *Prototype Development (2.6)*, and conducting a formal *Design Review & Approval (2.7)*.

3.0 Development (Core Operational Phase)

The core technical phase where system features, algorithms, and application interfaces are implemented.

- **3.1 Conversational AI Chatbot (Ask):** Structured into *NLP & Intent Design (3.1.1)* for user prompt understanding, *Bot Response Flow (3.1.2)* for logical dialogue management, and *Knowledge Base Integration (3.1.3)* for fetching accurate university data.
- **3.2 Smart Eligibility Matcher (Explore):** Comprises the *Eligibility Rule Engine (3.2.1)* for filtering background rules, *Student Profile Matching (3.2.2)* using student GPA inputs, and *Recommended Results Generation (3.2.3)*.
- **3.3 Side-by-Side Comparison (Choose):** Divided into *Comparison Logic (3.3.1)*, *Visualization (Charts/Tables) (3.3.2)* for side-by-side metric views, and dynamic *Filters & Sorting (3.3.3)*.
- **3.4 Notice & Deadline Tracker:** Encompasses *Data Collection (3.4.1)*, automated *Database Update Module (3.4.2)*, and a *Notification & Alert System (3.4.3)* for critical dates.
- **3.5 Frontend Development (Web UI):** Implements *Pages & Components (3.5.1)*, ensures cross-device *Responsive Design (3.5.2)*, and manages client-side *State Management (3.5.3)*.
- **3.6 Backend Development (APIs):** Handles *RESTful API Development (3.6.1)*, core *Business Logic Implementation (3.6.2)*, and external *Third-Party Integrations (3.6.3)*.
- **3.7 Module Integration:** Unifies frontend components, backend APIs, database models, and AI engine services into a single cohesive system.

4.0 Testing

Ensures system stability, accurate matching calculations, AI response verification, and secure deployment readiness.

- **Quality Assurance Workflows:** Begins with *Test Plan Preparation (4.1)* followed by multi-tier testing including *Unit Testing (4.2)*, *Integration Testing (4.3)*, and end-to-end *System Testing (4.4)*.
- **Validation & Optimization:** Evaluates real-world usability through *User Acceptance Testing (UAT) (4.5)*, alongside *Performance Testing (4.6)* and *Security Testing (4.7)*, concluding with iterative *Bug Fixing & Retesting (4.8)*.

5.0 Deployment & Documentation

Manages production go-live, documentation handovers, and final project defense.

- **Release Pipeline:** Involves *Environment Setup (5.1)*, production hosting via *Deployment & Hosting (5.2)*, and *Data Migration (If Required) (5.3)*.
- **Knowledge Transfer:** Provides *User Training & Guidelines (5.4)* and comprehensive *Documentation (5.5)* containing a *User Manual (5.5.1)*, *Technical Documentation (5.5.2)*, and *API Documentation (5.5.3)*, ending with the *Final Review & Project Handover (5.6)*.

□ GANTT CHART / PROJECT TIMELINE

The UniQuery project is scheduled across a **14-week academic timeline** divided into 5 major execution phases. Detailed task dependencies, owner allocations, and weekly schedules are illustrated in the project Gantt Chart below.



Figure: A 14-week timeline visualization showing task schedules, dependencies, owner roles, and target milestone dates.

Timeline & Phase Overview

- **Phase 1: Planning & Initiation (Weeks 1–4):** Project charter approval, requirements gathering, and feasibility study conducted jointly by the team.
- **Phase 2: System Design (Weeks 5–8):** Drafting UML models, system architecture, database ER diagrams, and UI/UX wireframes.
- **Phase 3: Core Development (Weeks 5–12):** Parallel feature engineering—Israt Jahan (Member 1) handles the comparison dashboard, notice tracker, and web UI; Md. Mehedi Hasan (Member 2) builds the AI chatbot, eligibility matcher, and backend REST APIs, followed by full system integration.
- **Phase 4: Testing & Quality Assurance (Weeks 10–13):** Executing functional testing, User Acceptance Testing (UAT), performance checks, and iterative bug fixes.
- **Phase 5: Deployment & Handover (Weeks 13–14):** Production cloud deployment, user/technical documentation, and final project defense.

Key Project Milestones

- **Milestone 1 (Week 1):** Project Charter Approved
- **Milestone 2 (Week 4):** System Requirements Specification Completed
- **Milestone 3 (Week 7):** System Architecture & Database Design Finalized
- **Milestone 4 (Week 12):** Core Feature Development & Integration Completed
- **Milestone 5 (Week 13):** System Testing & UAT Sign-Off
- **Milestone 6 (Week 14):** Production Go-Live & Final Project Handover

□ BUDGET / COST ESTIMATION

The budget for the UniQuery academic software project is derived from current market pricing research in Bangladesh as of 2026.

Resource Categories & Free vs. Paid Allocations

- **Free / Open-Source Resources:**
 - *Development Hardware:* Existing personal student laptops (0 BDT).
 - *IDE & Tooling:* VS Code, Git, GitHub, Postman, Figma Free Plan (0 BDT).
 - *Frontend Deployment:* Vercel Hobby Tier provides free continuous integration and SSL certificates for web applications (0 BDT).
 - *Backend Service Hosting:* Render Free Web Service Tier provides sufficient cloud execution runtime for REST APIs (0 BDT).
 - *Database Hosting:* Supabase Free PostgreSQL Tier provides 500 MB relational database storage, which easily satisfies academic project requirements (0 BDT).
- **Paid Resources:**
 - *Custom Domain Name:* 1-year .com domain registration for official public accessibility.
 - *AI / LLM API Usage:* OpenAI gpt-4o-mini API tier for conversational AI chatbot queries.

Variable Cost Calculations & Usage Assumptions

1. **AI Chatbot API Token Usage Assumptions:**
 - Target prototype testing volume: 1,000 queries/day across 60 days of development and testing = 60,000 total conversation turns.
 - Average token footprint per conversation turn: 500 input tokens (system instructions + database context + student prompt) and 300 output tokens (generated chatbot response).
 - Total Input Tokens: $60,000 \times 500 = 30,000,000$ tokens (30M).

- Total Output Tokens: $60,000 \times 300 = 18,000,000$ tokens (18M).
- Unit Rates ([gpt-4o-mini](#)): \$0.15 / 1 Million Input Tokens; \$0.60 / 1 Million Output Tokens.
- Input Cost: $30M \times \$0.15 = \4.50 USD.
- Output Cost: $18M \times \$0.60 = \10.80 USD.
- Total Estimated API Cost: \$15.30 USD $\approx$ BDT 1,850 (calculated at 1 USD = 121 BDT).

2. Domain Name Registration:

- 1-year [.com](#) TLD registration via local/international vendor: BDT 499.

Comprehensive Cost Estimation Table

No.	Resource / Service	Plan / Specification	Quantity/ Duration	Unit Cost	Estimated Cost (BDT)	Market Source & Justification
1	Custom Domain	.com TLD Registration	1 Domain / 1 Year	BDT 499 / Year	BDT 499	Source: Hostinger BD (hostinger.com/bd/tld/com-domain) checked 2026. Essential for platform deployment.
2	Frontend Hosting	Vercel Hobby Tier	1 App / 1 Year	BDT 0 (Free)	BDT 0	Source: Vercel Official (vercel.com/pricing). Free static/SSR hosting with built-in SSL.
3	Backend Hosting	Render Free Web Service	1 Service / 1 Year	BDT 0 (Free)	BDT 0	Source: Render Official (render.com/pricing). Sufficient for prototype API backend.

4	Cloud Database	Supabase PostgreSQL Free	1 DB / 1 Year	BDT 0 (Free)	BDT 0	Source: Supabase Pricing (supabase.com/pricing). 500MB DB capacity is ample for university records.
5	AI Chatbot API	OpenAI API (gpt-4o-mini)	60M Tokens Total	\$0.15 / \$0.60 per 1M	BDT 1,850	Source: OpenAI Pricing (openai.com/api/pricing). Covers ~60k testing queries for conversational bot.
6	Development IDE	VS Code / Git / Postman	Permanent	BDT 0 (Open Source)	BDT 0	Standard student development environment.
7	Hardware Setup	Student Laptops	2 Devices	BDT 0 (Existing)	BDT 0	Existing hardware utilized; no new hardware purchased.
-	TOTAL ESTIMATED COST				BDT 2,349	(Approx. ~\$19.50 USD)

❑ EXPECTED OUTCOMES

- **Centralized Information Hub:** A functioning web platform unifying public and

private university admission circulars, unit criteria, and program costs.

- **Reduced Guidance Overhead:** Instant conversational answers for applicants, reducing manual research time.
- **Accurate Eligibility Determination:** Instant profile matching preventing applicants from applying for ineligible programs.
- **Transparent Financial & Campus Comparisons:** Side-by-side matrices enabling students and parents to evaluate costs and available waiver tiers.
- **Decreased Missed Deadlines:** Clear notice boards keeping students informed of critical application and exam dates.

❑ LIMITATIONS

- **Data Scope Boundary:** Prototype database will be populated with a representative subset of prominent universities rather than all nationwide institutions.
- **Static Circular Data Dependency:** Relies on stored administrative update pipelines rather than live official university API feeds.
- **API Rate & Network Dependencies:** System responsiveness relies on active third-party LLM API availability and internet connectivity.
- **Third-Party Application Process:** System directs users to external official portals for final application submissions.

❑ FUTURE SCOPE

- **Comprehensive Nationwide Expansion:** Expanding the database to cover all accredited public, private, and national university colleges in Bangladesh.
- **Push Notification Engine:** Integrating SMS and email notifications for personalized deadline alerts and circular releases.
- **Mobile Application Build:** Developing cross-platform mobile apps (Flutter/React Native) for Android and iOS devices.
- **Multilingual Support:** Implementing dynamic Bangla and English language toggles within the conversational AI chatbot.
- **Direct Application Gateway:** Partnering with institutions to support unified in-app application form submissions.

❏ CONCLUSION

Selecting a university is a critical, complex decision frequently hindered by scattered information, obscure eligibility criteria, and strict deadlines. **UniQuery** solves this challenge through its **"Ask, Explore, Choose"** framework, combining a conversational AI chatbot, automated eligibility checking, and structured visual comparisons. Built by a dedicated 2-member team, the platform is technically achievable, economically feasible, and tailored to provide clear academic value.